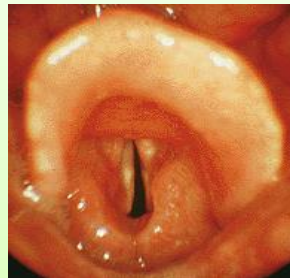


CARTILAGES, MEMBRANES, MUSCLES AND THE MUCOSA OF THE LARYNX



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SEMMELWEIS UNIVERSITY

Department of Anatomy, Histology and Embryology

POSITION

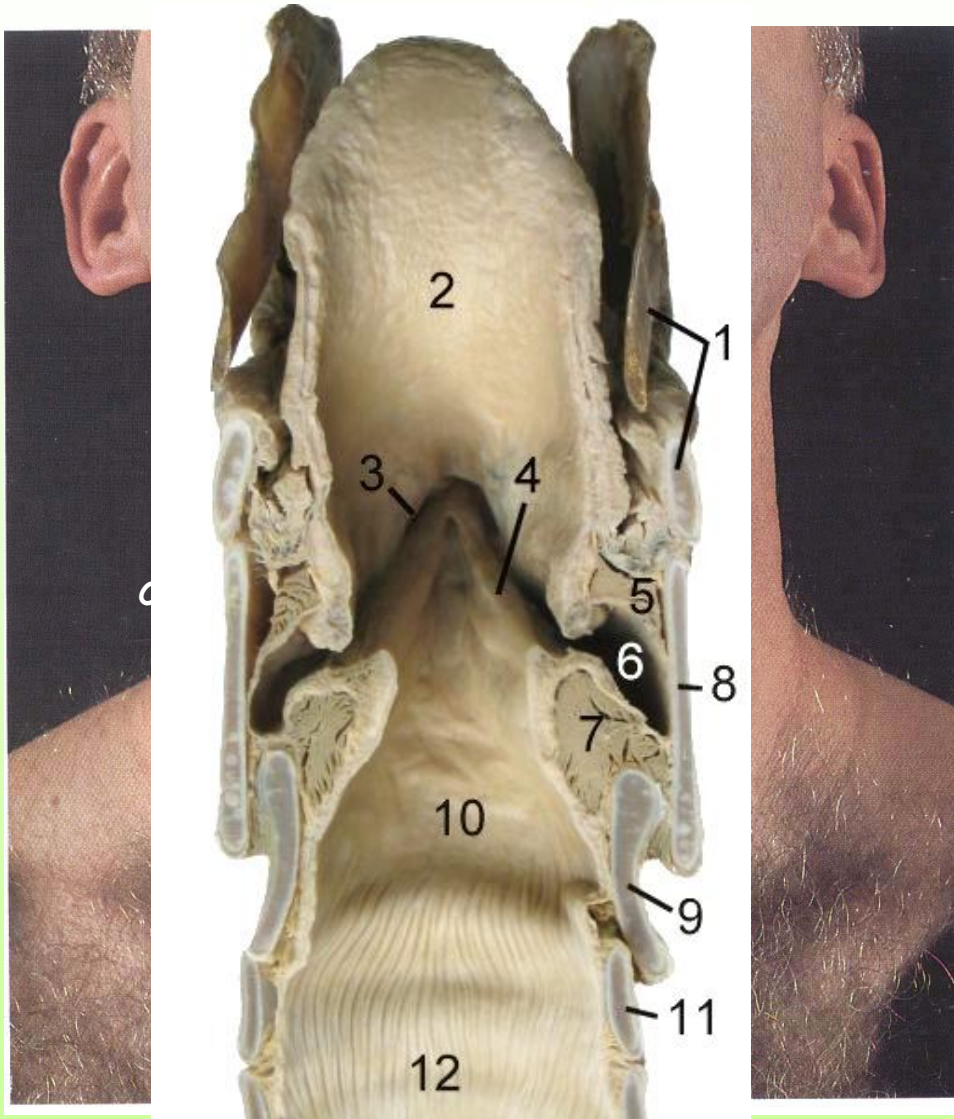
LARYNGEAL „HANDSHAKE”



WHAT ARE YOU GOING TO LEARN ABOUT?

- Position of the larynx - diagnostic devices (what is visible)
- Phonation and respiration
- Bones, cartilages, membranes, ligaments
- Internal lining – mucosal folds AND spaces
- Innervation and blood supply
- Joints, muscles, actions

THE LARYNX



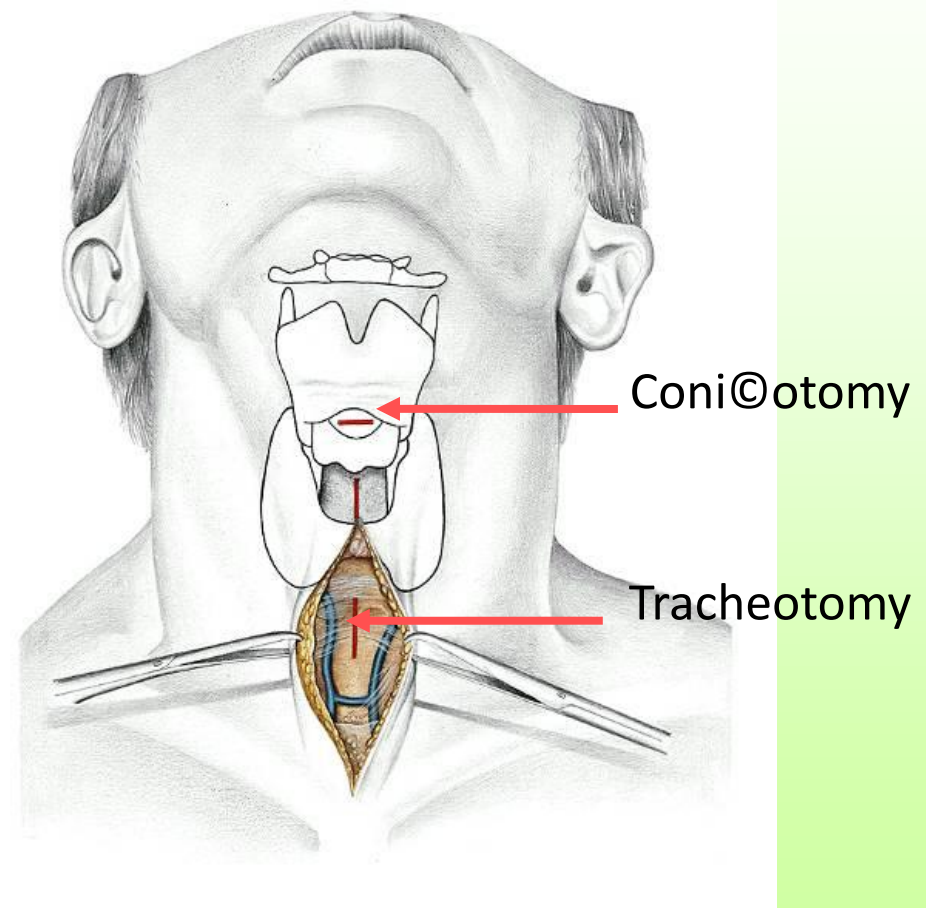
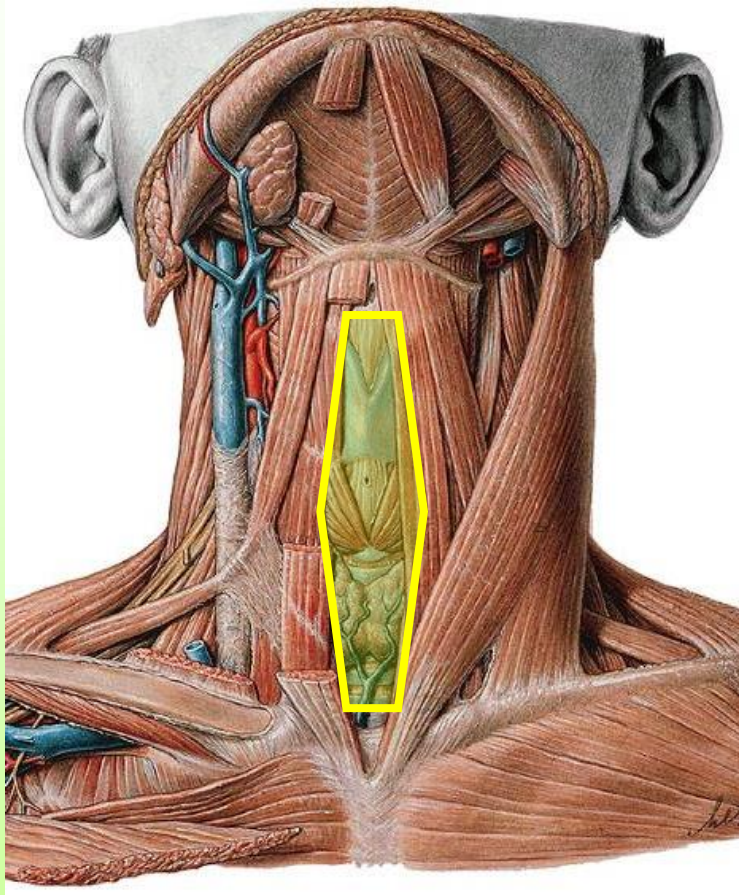
MAJOR FUNCTIONS

- To separate the respiratory and digestive tracts
- Phonation
- respiration
- Cough
- Abdominal crunch

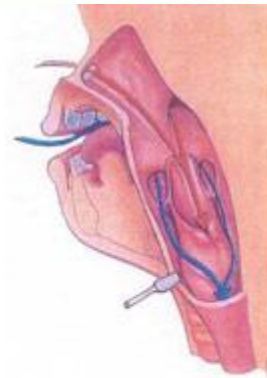
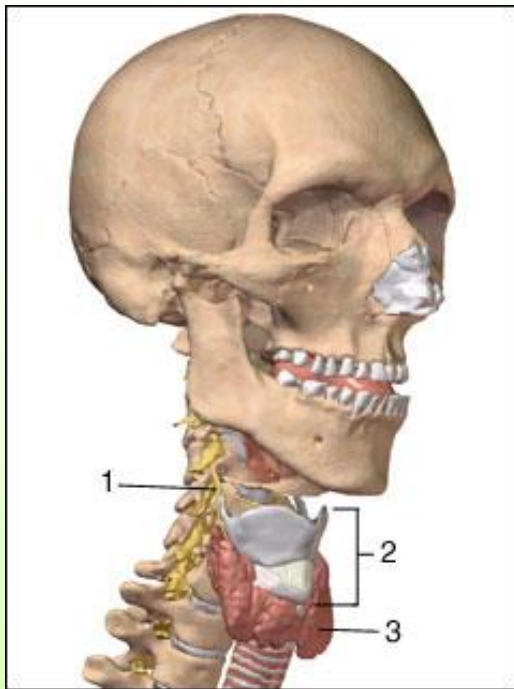
CLINICAL RELEVANCES

- Loose Submucosa (Glottic edema)
- Laryngitis
- Croup (Diphtheria)
- Pseudocroup (subglottic laryngitis)
- Oncological diseases
- Conicotomy, Intubation

MEDIAN CERVICAL REGION



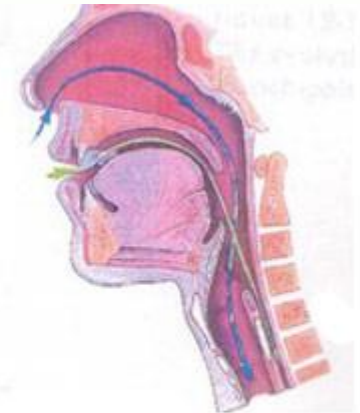
DESCENT OF THE LARYNX (DEVELOPMENT)



oral cavity
of an adult
chimpanzee



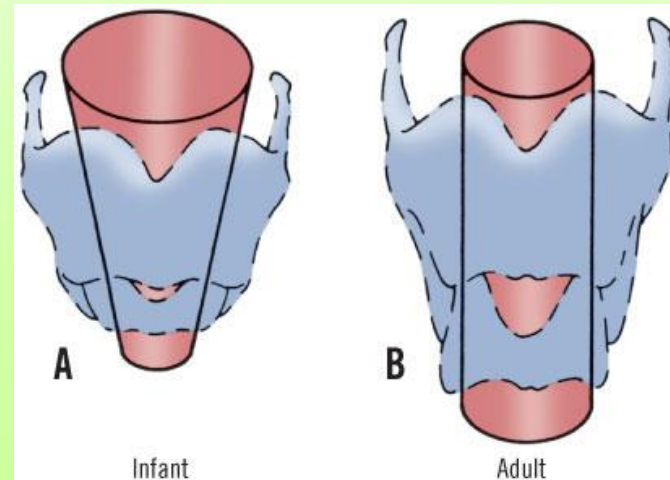
oral cavity of a
baby less than
9 months old



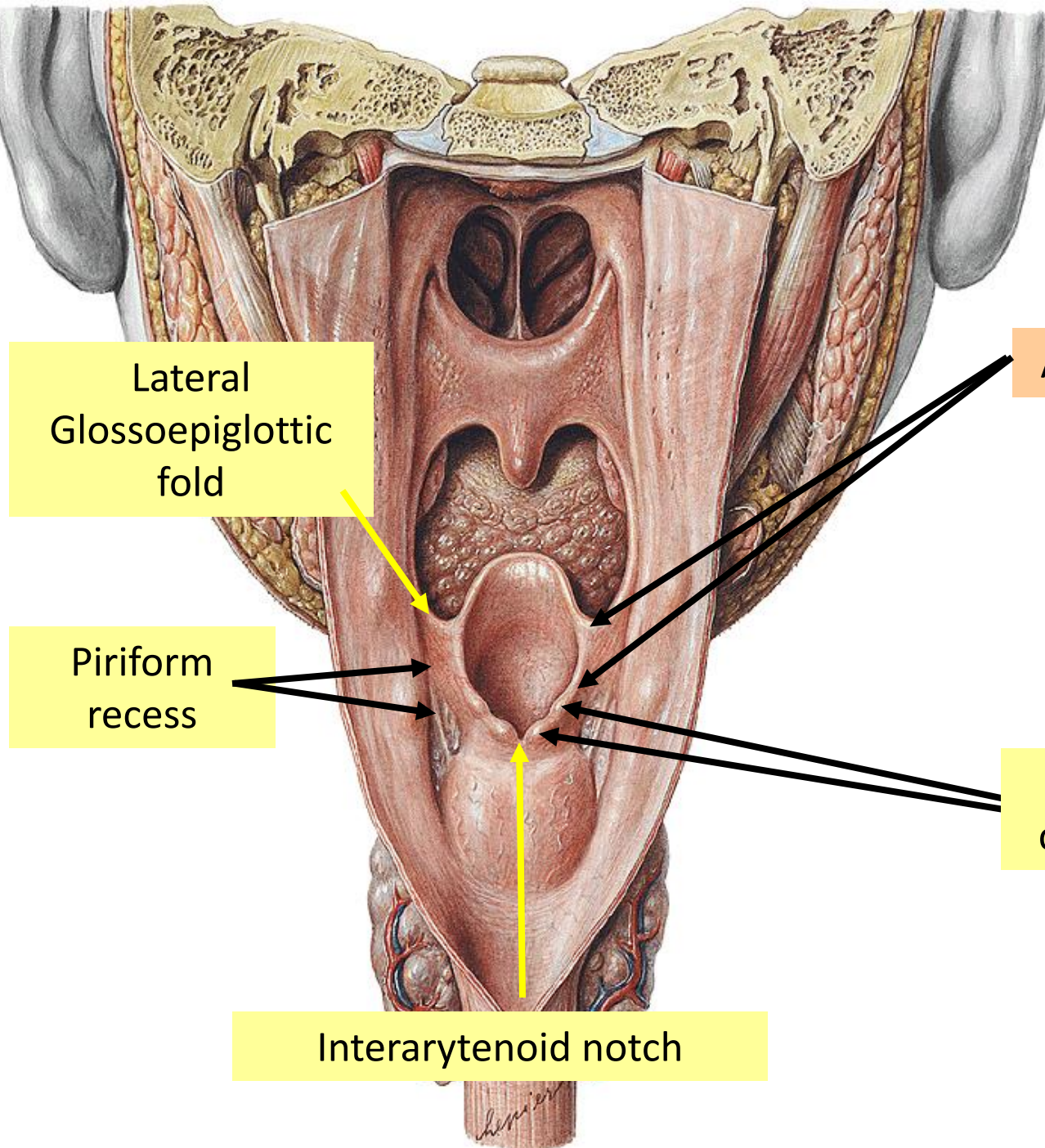
oral cavity and
larynx of an
adult man

Newborn babies C 3 - 4

Adults C 4 - 6
upper edge of thyroid C4/5,
cricoid cartilage C6



ADITUS LARYNGIS



Lateral
Glossoepiglottic
fold

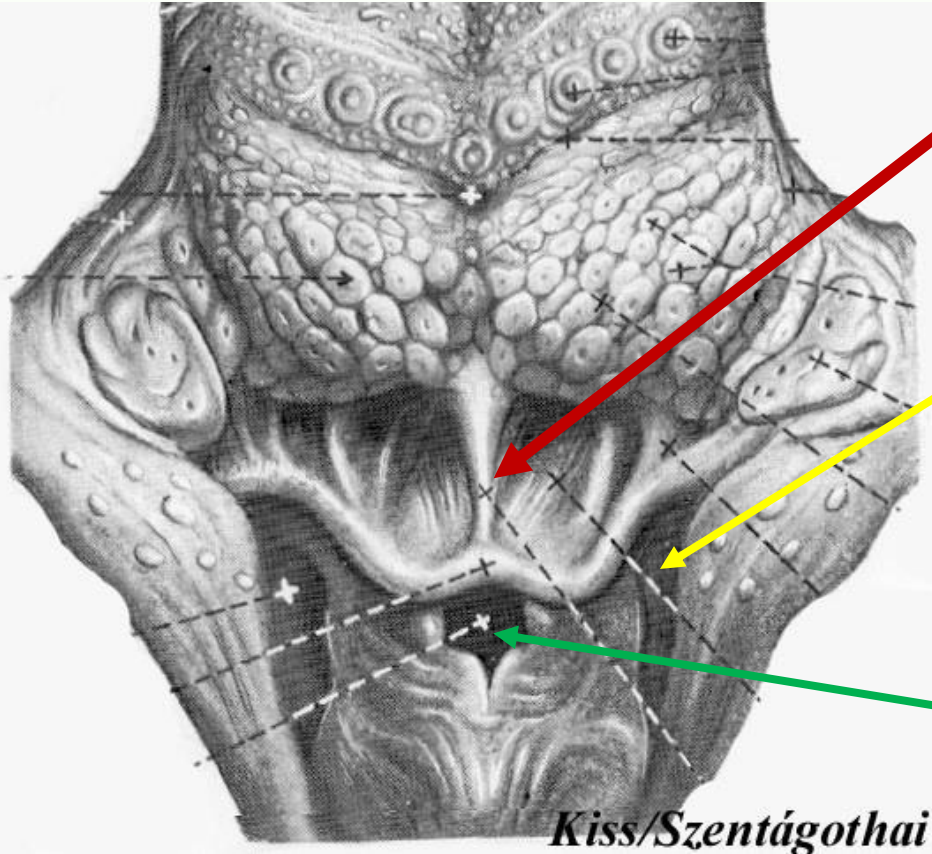
Piriform
recess

Aryepiglottic fold

Corniculate and
cuneiform tubercles

Interarytenoid notch

MUCOUS LINING OF THE CAVITIES



EPIGLOTTIS

Glossoepiglottic folds
Epiglottic Valleculae

PIRIFORM RECESS

*Fold of the superior laryngeal
nerve*

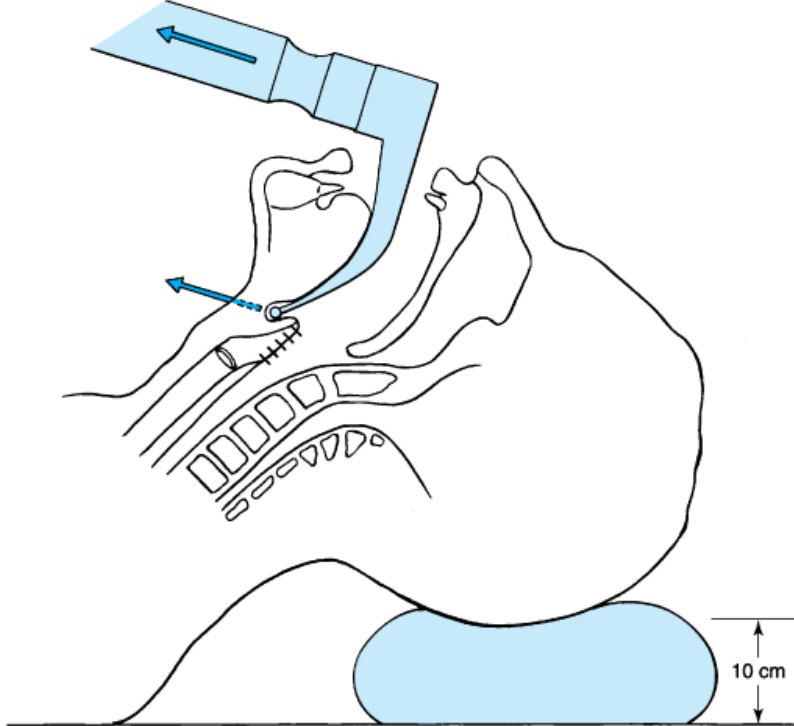
ENTRANCE TO THE LARYNX

ADITUS LARYNGIS (BORDERS)

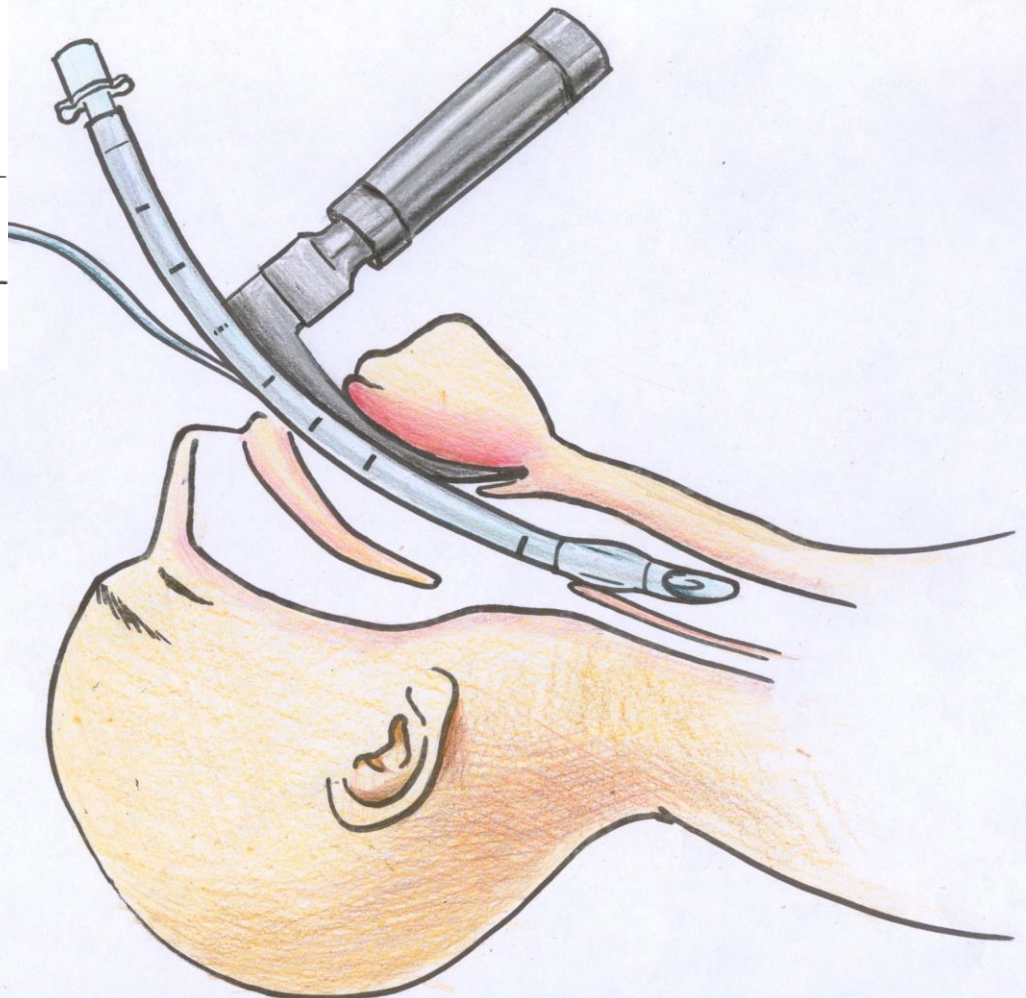
Aryepiglottic fold
Cuneiform tubercle (Wrisberg)
Corniculate tubercle (Santorinus)
Interarytenoid notch

Stratified nonkeratinizing squamous epithelium vs. Respiratory epithelium

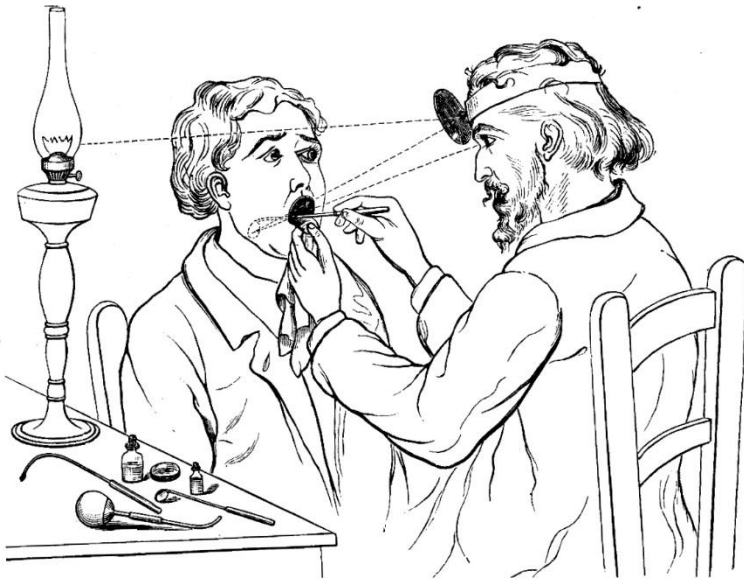
INTUBATION FOR GENERAL ANAESTHESIA



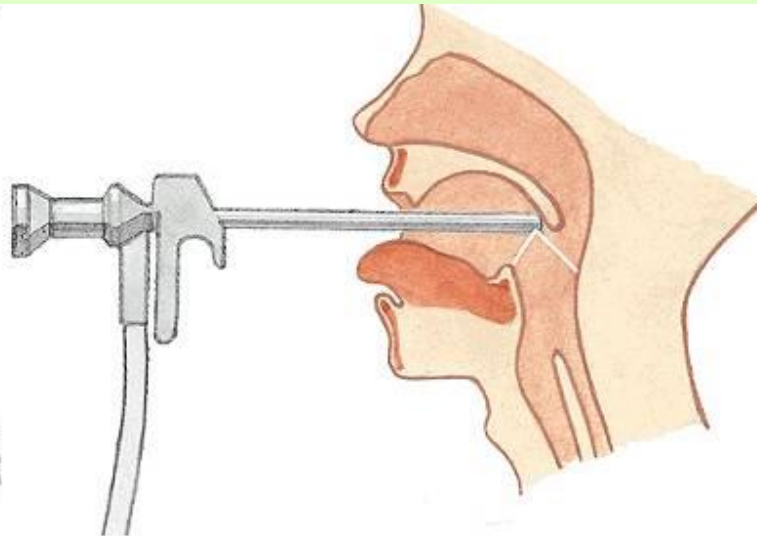
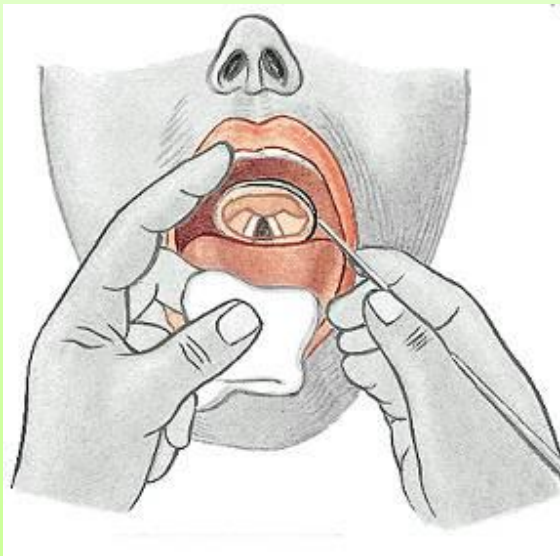
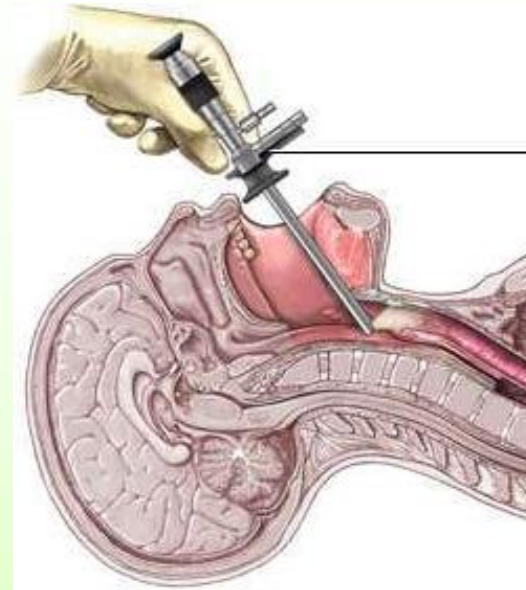
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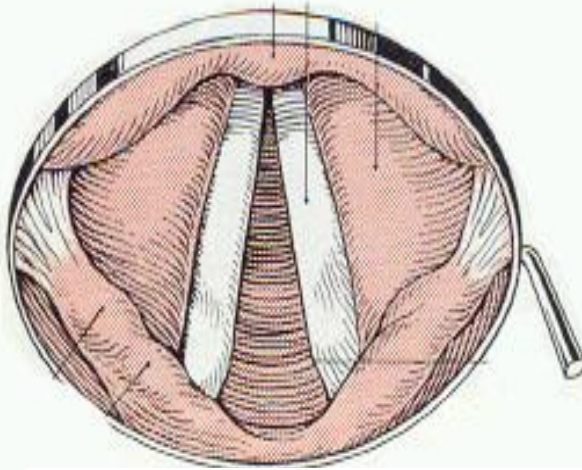
LARYNGOSCOPY TO INSPECT THE ORGAN



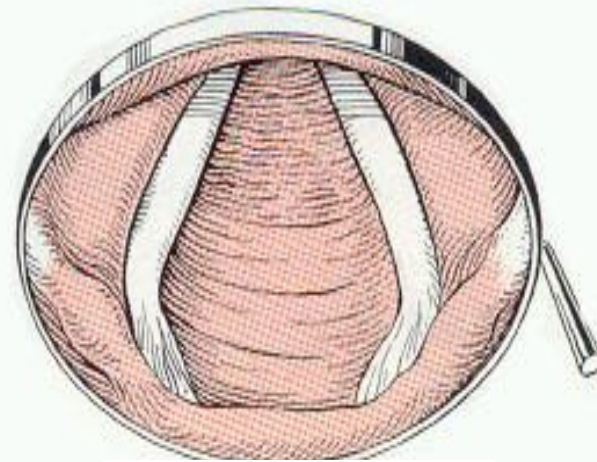
Situationsbild bei der laryngoskopischen Untersuchung.



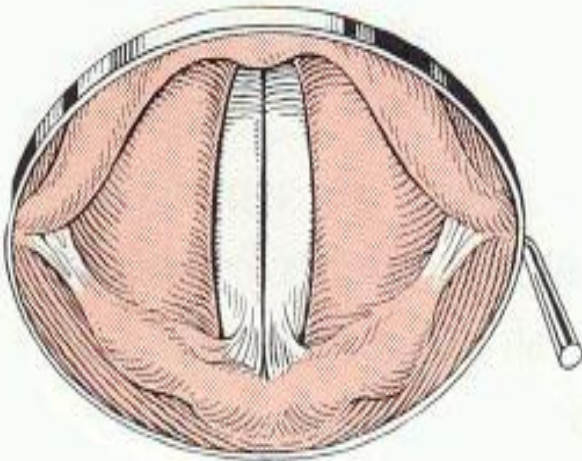
WHAT CAN YOU SEE IN THE LARYNGOSCOPE?



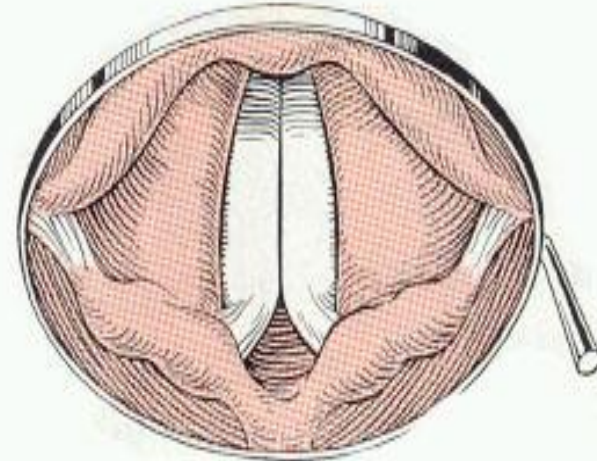
Middle respiratory position



Strong respiration



Phonation (closure of Glottis)



whisper

PHONATION, VOCALIZATION

air is expelled from the lungs through the glottis

Causing **pressure drop** across the larynx
(Threshold 2-3 H_2O cm)

the vocal folds start to **oscillate** (mostly laterally)

The oscillation **modulates** the pressure and flow of the air through the larynx

SOUND = HARMONIC SERIES

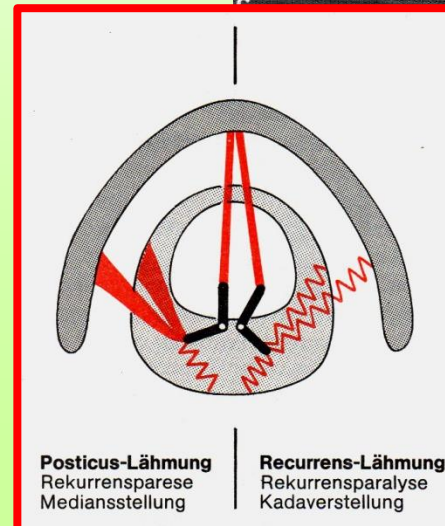
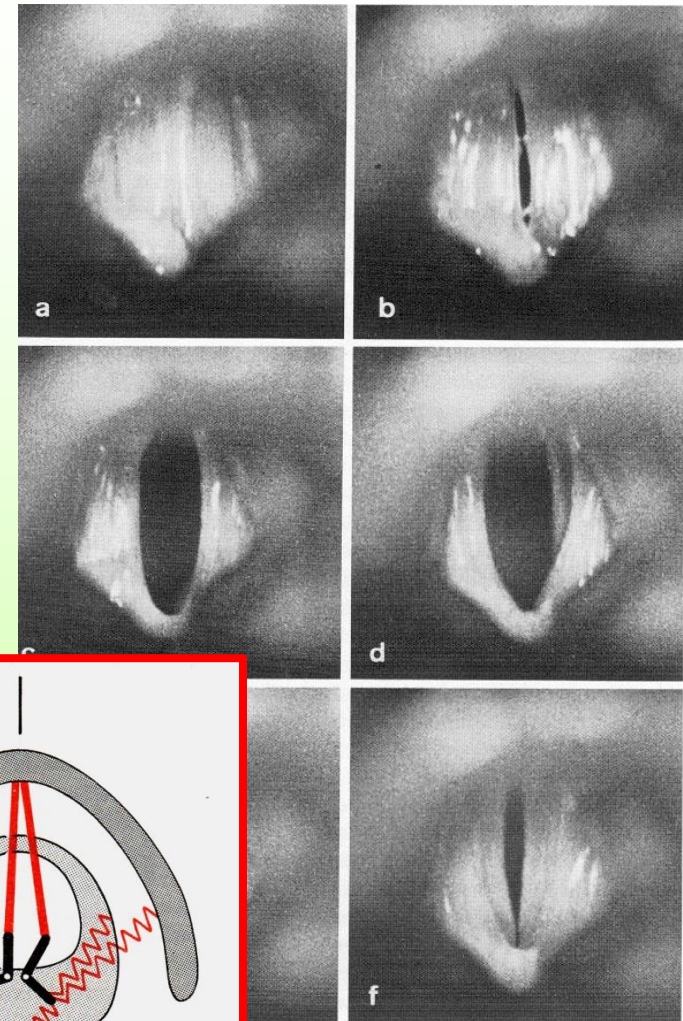
It consists of a **fundamental tone** (or fundamental frequency, the main acoustic cue for the percept pitch) accompanied by **harmonic overtones**, which are multiples of the fundamental frequency.

It can be varied by

- Pressure changes (lung capacity)
- Muscles large scale: **CRICOTHYROID**.
minor changes: **THYROARYTENOID**
- Positional changes (due to atm to the hyoid bone)

OR

PARALYSIS

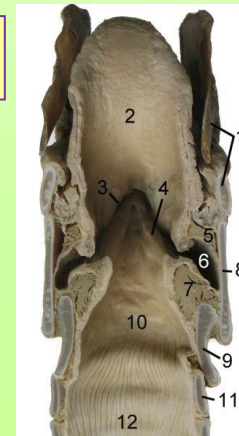
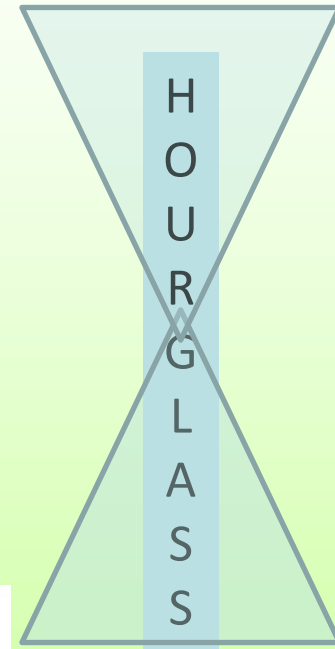
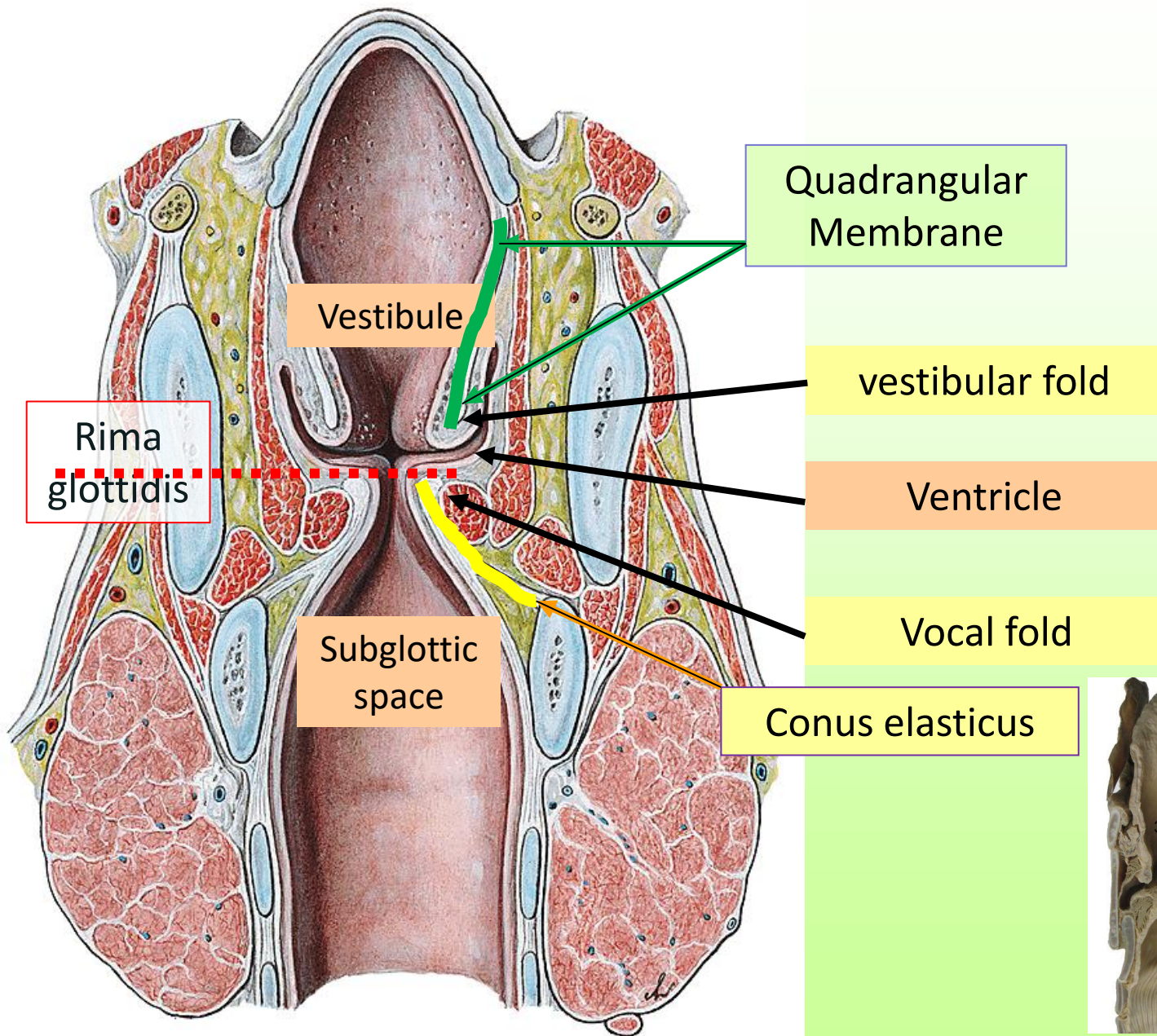


Posticus-Lähmung
Rekurrensparese
Mediansstellung

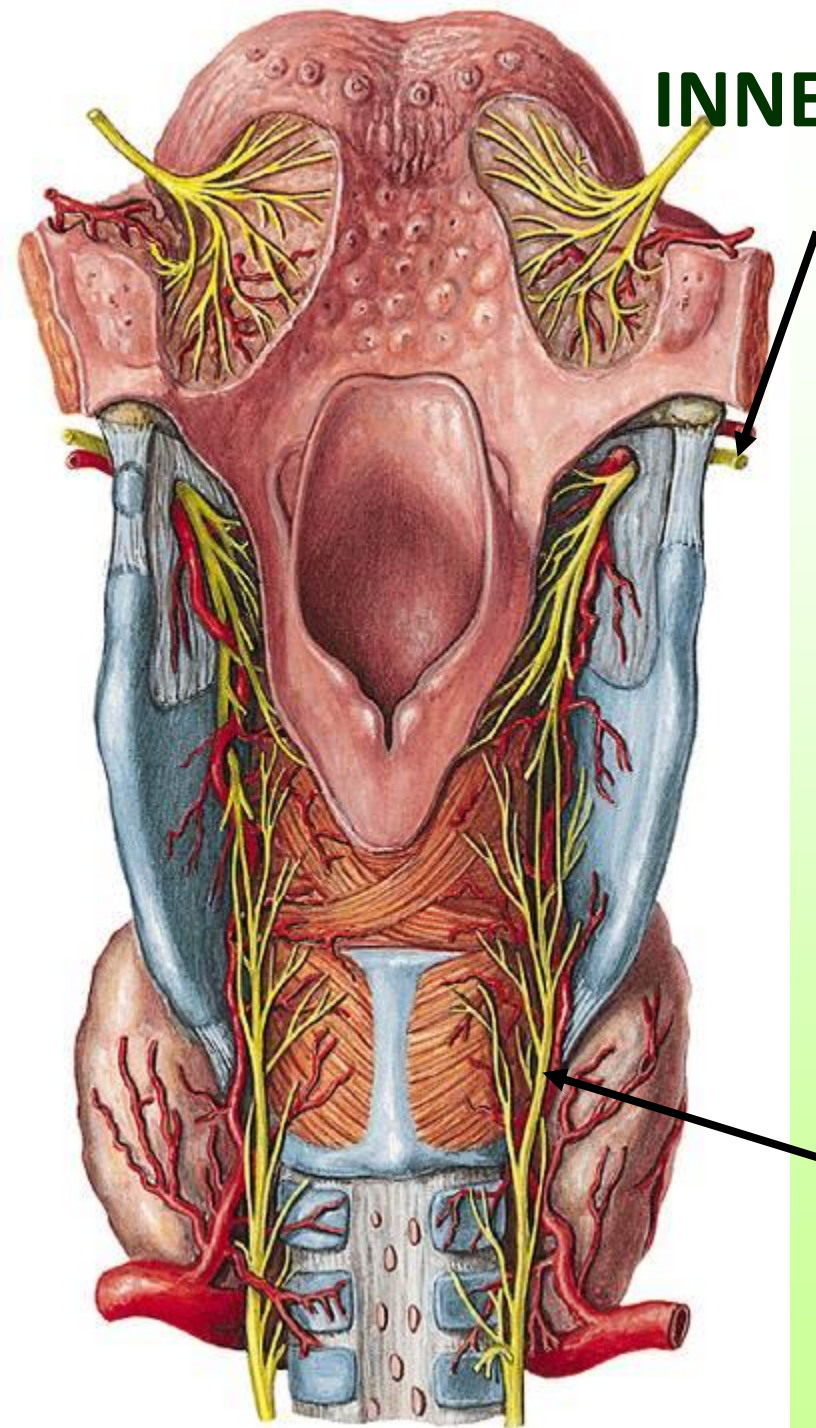
Recurrans-Lähmung
Rekurrensparese
Kadaverstellung



SPACES WITHIN THE LARYNX



INNERVATION AND BLOOD SUPPLY



superior laryngeal nerve (CN 10.)

- cricothyroid
- Mucosa above the rima

Superior laryngeal artery **EXTERNAL CAROTID**
(branch of the sup. thyroid)

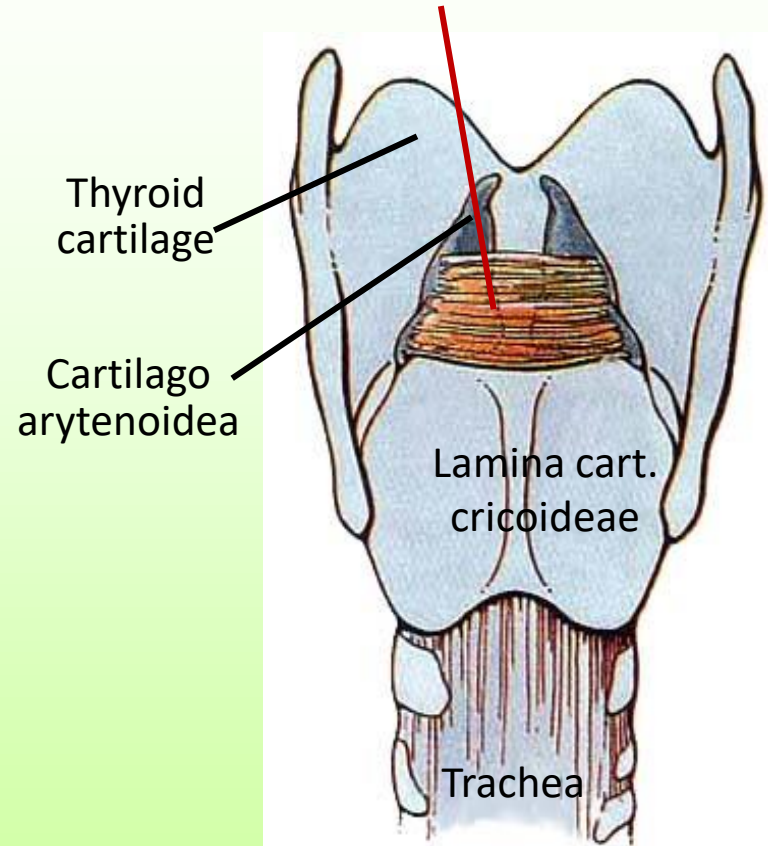
Inferior laryngeal artery **SUBCLAVIAN**
(branch of the inf. thyroid)

inferior (recurrent) laryngeal nerve (CN 10.)

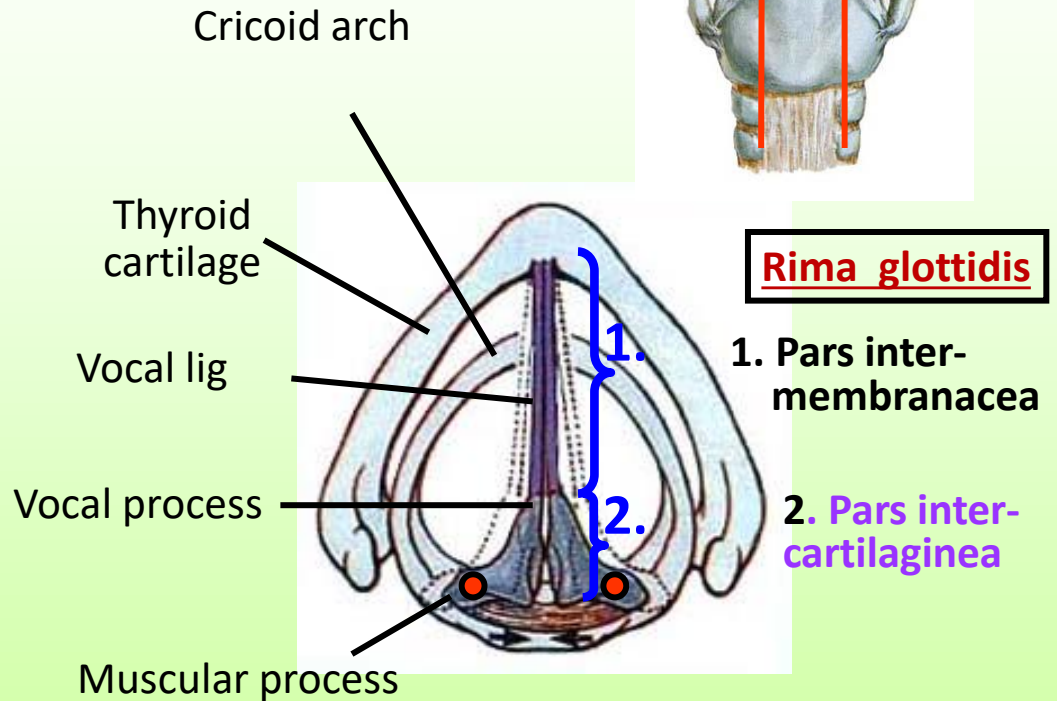
- all internal muscles
- Mucosa below the rima

LARYNGEAL MUSCLES

(INTER)ARYTENOID MUSCLE TRANSVERSE PART



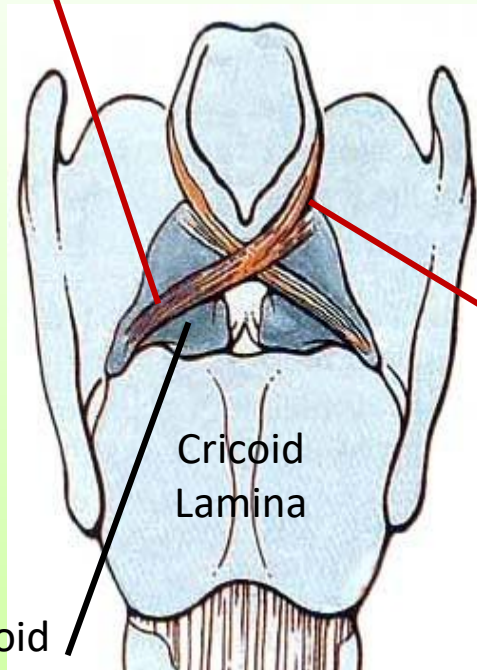
ORIGIN: } On the back of the arytenoid
INSERTION: } (the muscle lies internal to the vertical
axis of the cricoarytenoid joint



CLOSES THE RIMA GLOTTIDIS
At the pars intercartilaginea

LARYNGEAL MUSCLES

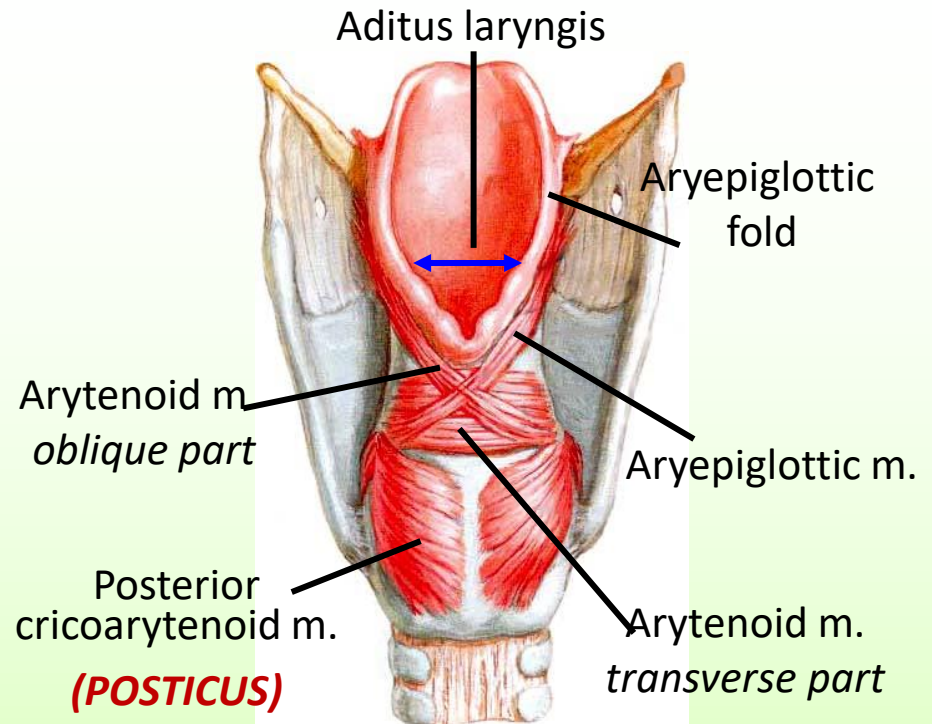
(INTER)ARYTENOID MUSCLE *OBLIQUE PART*



Arytenoid
cartilage

Cricoid
Lamina

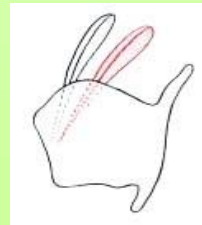
CLOSES THE RIMA GLOTTIDIS
at the **Pars intercartilaginea**



ARYEPIGLOTTIC MUSCLE

ORIGIN continuation of the obl. arytenoid muscle.

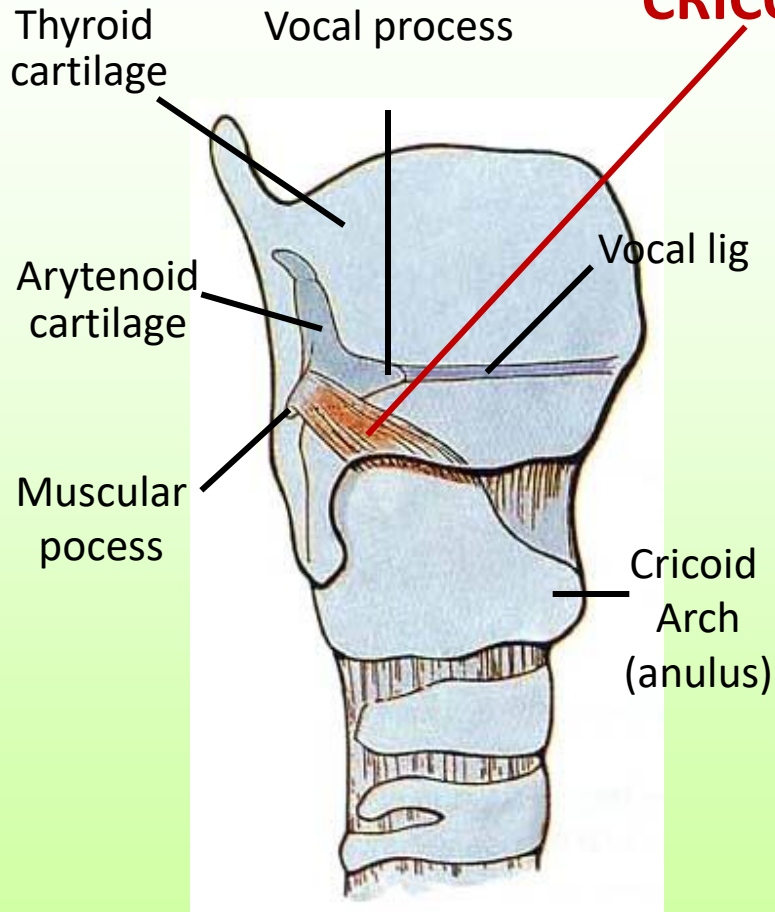
INSERTION aryepiglottic fold



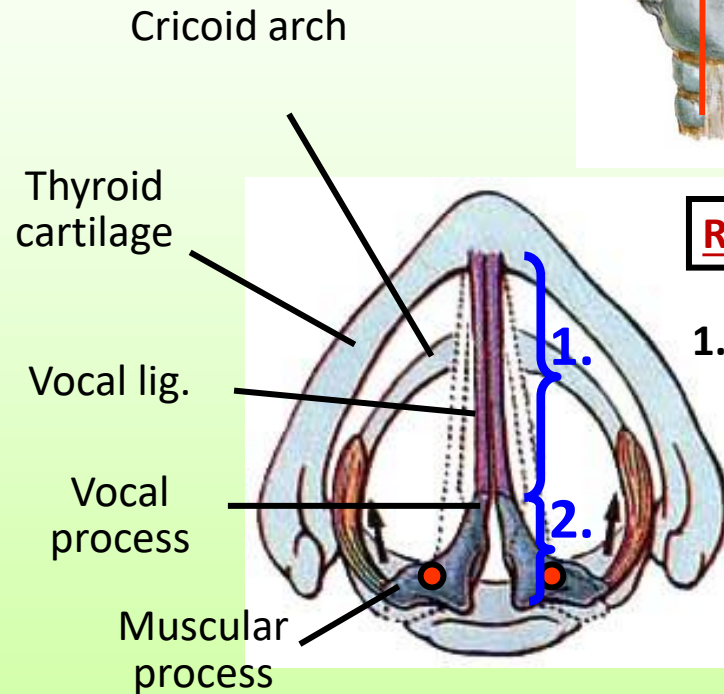
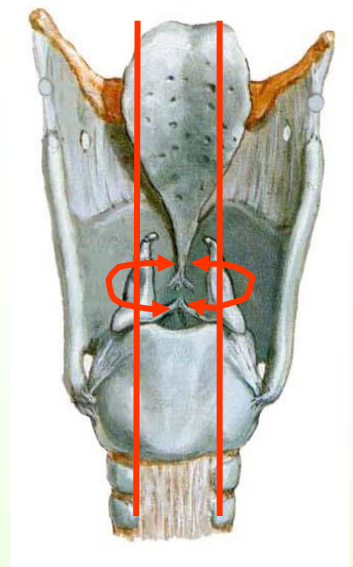
Pulls down the epiglottis,
tightening the larynx

LARYNGEAL MUSCLES

LATERAL CRICOARYTENOID MUSCLE



ORIGIN cricoid arch (upper lateral edge)
INSERTION muscular proc. of arytenoid



Rima glottidis

1. Pars intermembranacea

2. Pars intercartilaginea

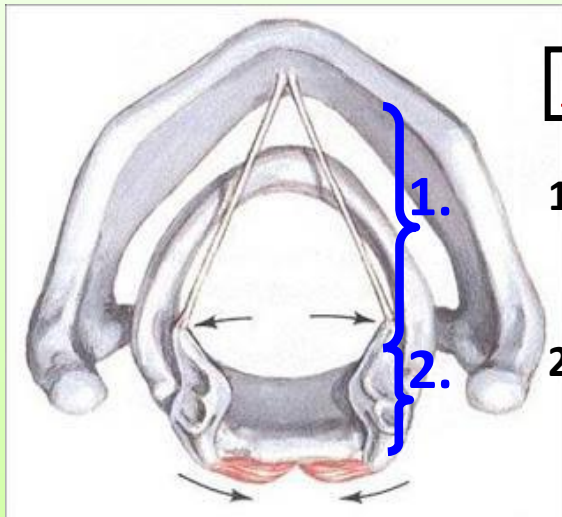
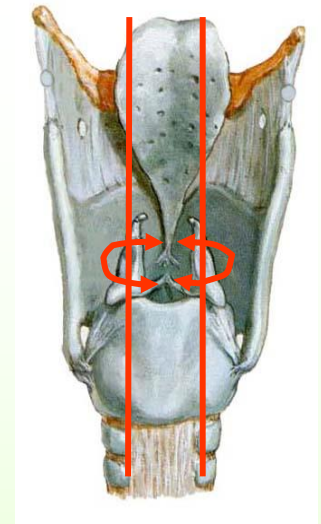
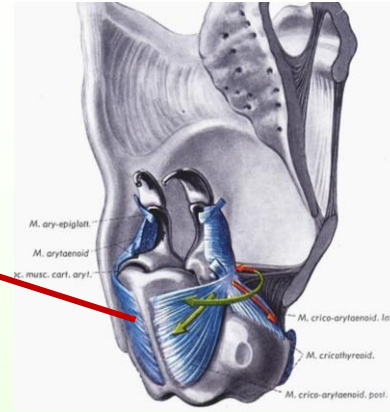
CLOSES THE RIMA GLOTTIDIS
At the pars intermembranacea

LARYNGEAL MUSCLES

POSTERIOR CRICOARYTENOID MUSCLE „POSTICUS”

ORIGIN cricoid lamina

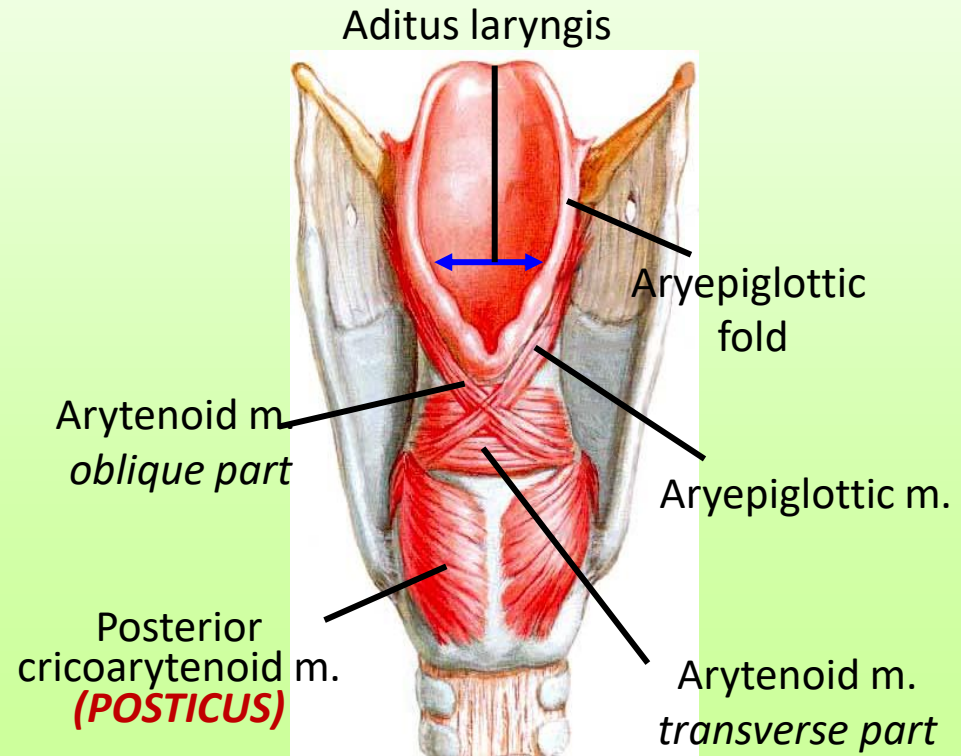
INSERTION muscular proc. of arytenoid



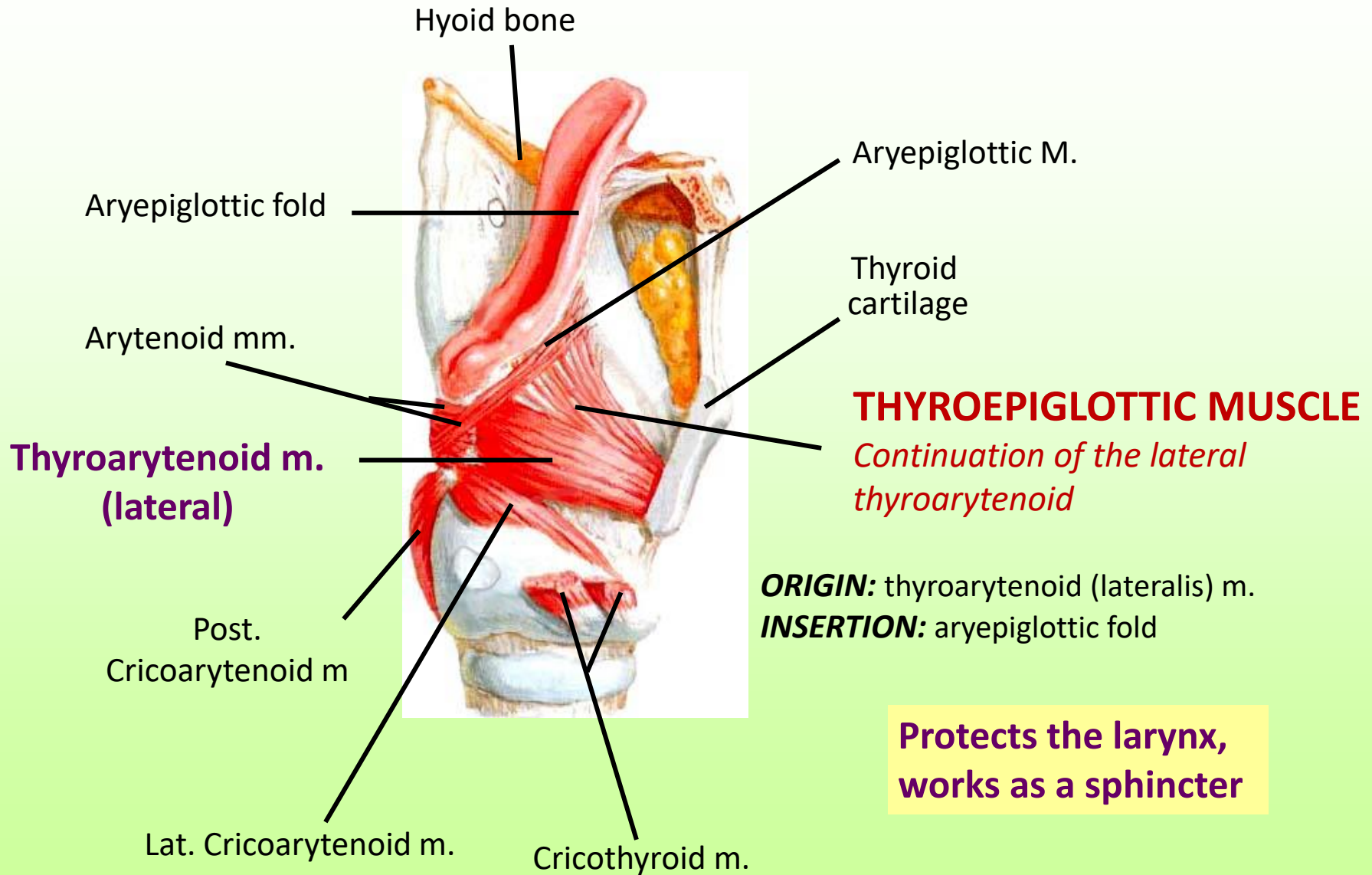
Rima glottidis

1. Pars intermembranacea
2. Pars intercartilaginea

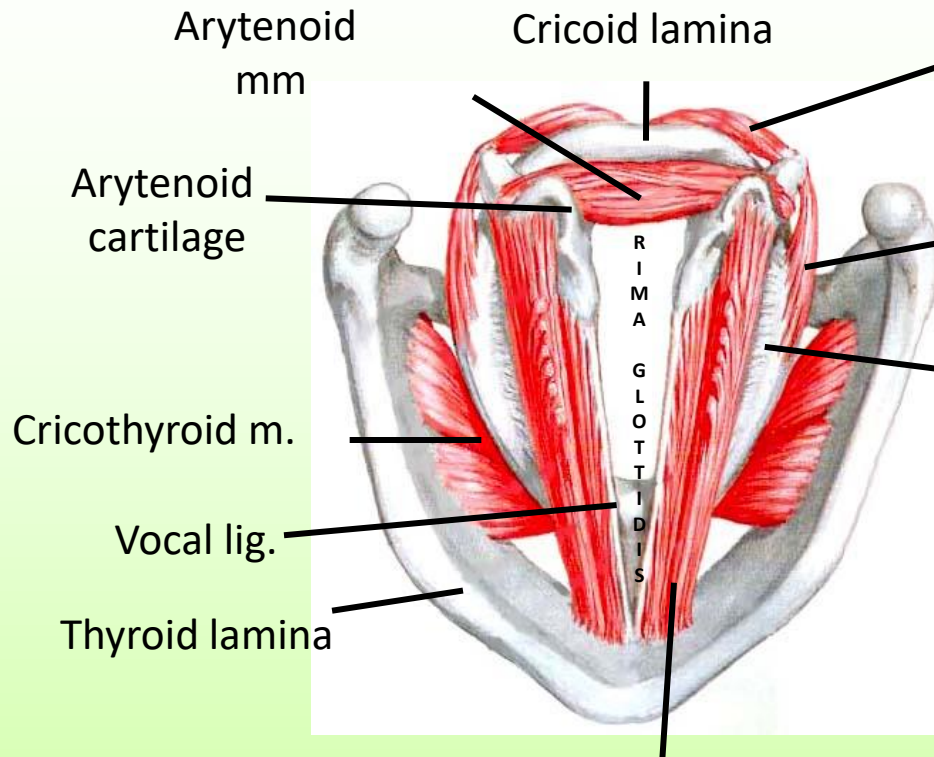
**THE ONLY OPENER OF THE
RIMA GLOTTIDIS**



LARYNGEAL MUSCLES



LARYNGEAL MUSCLES



Post. Cricoarytenoid muscle

Lat. Cricoarytenoid m.

LATERAL THYROARYTENOID MUSCLE

ORIGIN: lateral rim of the arytenoid cartilage

INSERTION: post aspect of the plate of thyroid

**Shortens and relaxes the vocal fold
CLOSES THE RIMA GLOTTIDIS**

VOCALIS MUSCLE (MEDIAL THYROARYTENOID)

**CLOSES THE RIMA GLOTTIDIS
At the pars intermembranacea**

ORIGIN: antero-lateral surface of the arytenoid cartilage

INSERTION: lamina cartilaginis thyroideae hintere Fläche

It fills the vocal fold

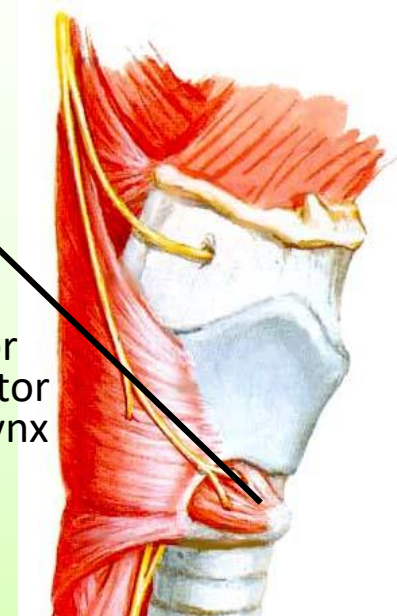
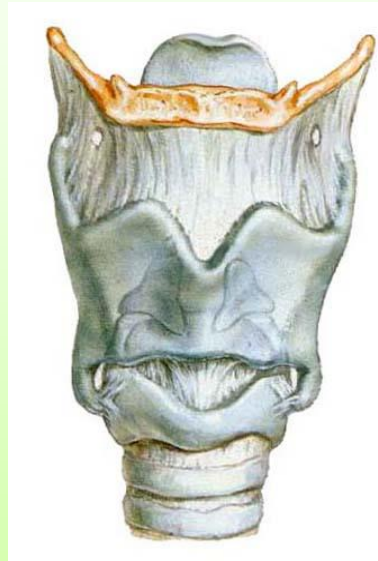
During contraction the vocal cord is stretched and vibrates „better“.

LARYNGEAL MUSCLES

WHICH MUSCLE HAVE WE MISSED?

the only tensor muscle of the larynx aiding with phonation

M. Cricothyroideus



ORIGIN + INSERTION:

splits into two groups or parts.

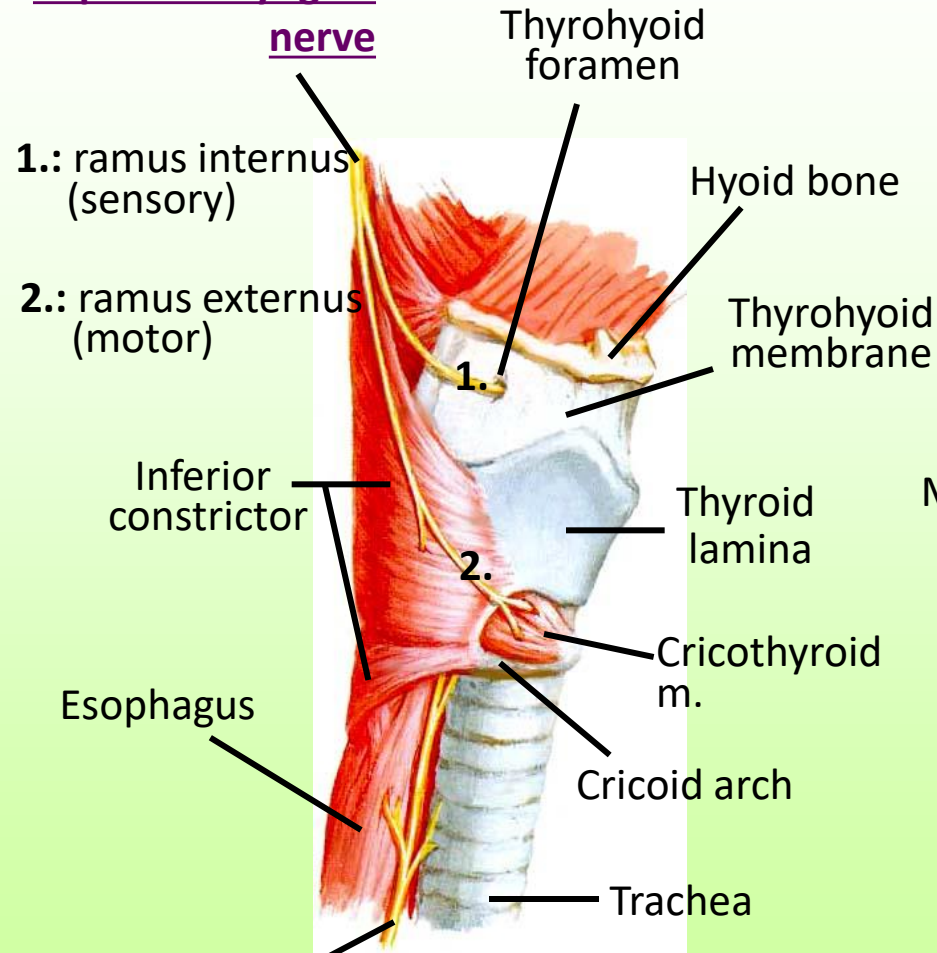
oblique part travels posterolaterally and inserts onto the inferior cornu of the thyroid cartilage.

straight part travels posterosuperiorly and inserts onto the inferior margin of the lamina of the thyroid cartilage

Tilts the thyroid around the transverse axis (stretches the vocal cord elevating the pitch of voice)

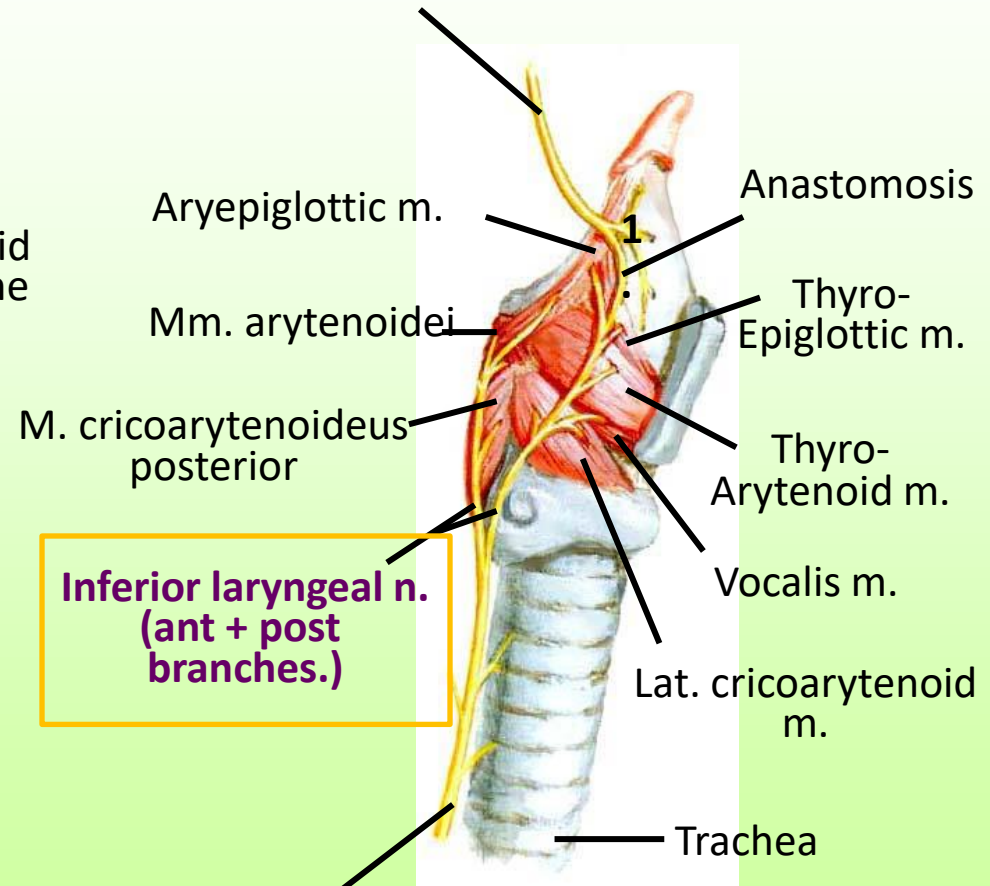
INNERVATION OF THE LARYNGEAL MUSCLES

Superior laryngeal nerve



Recurrent laryngeal nerve

Superior laryngeal nerve



**Inferior laryngeal n.
(ant + post
branches.)**

Recurrent laryngeal nerve

CLOSURE OF THE ADITUS DURING SWALLOWING

There is no complete closure

The pharyngeal constrictors contract, closing the **Aditus** from behind

The soft palate and the uvula are lifted to block the pharynx from above

Reflex CN 10.

NORMAL POSITION



SWALLOWING



ROOT OF THE TONGUE – EPIGLOTTIS

Mechanism

The muscles of the floor of the oral cavity (Mm. mylohyoideus, digastrici et thyrohyoidei)



The larynx is elevated



The aditus is approaching (lifted to) the epiglottis

ADIPOSE BODY – EPIGLOTTIS

Mechanism

The root of the tongue is driven posterior



The preepiglottic adipose tissue is pushed against the epiglottis



aryepiglottic and thyroepiglottic muscles contract to pull the larynx and the epiglottis closer

HISTOLOGY OF THE LARYNX

Laryngeal vestibule

VESTIBULAR FOLDS

Respiratory epithelium

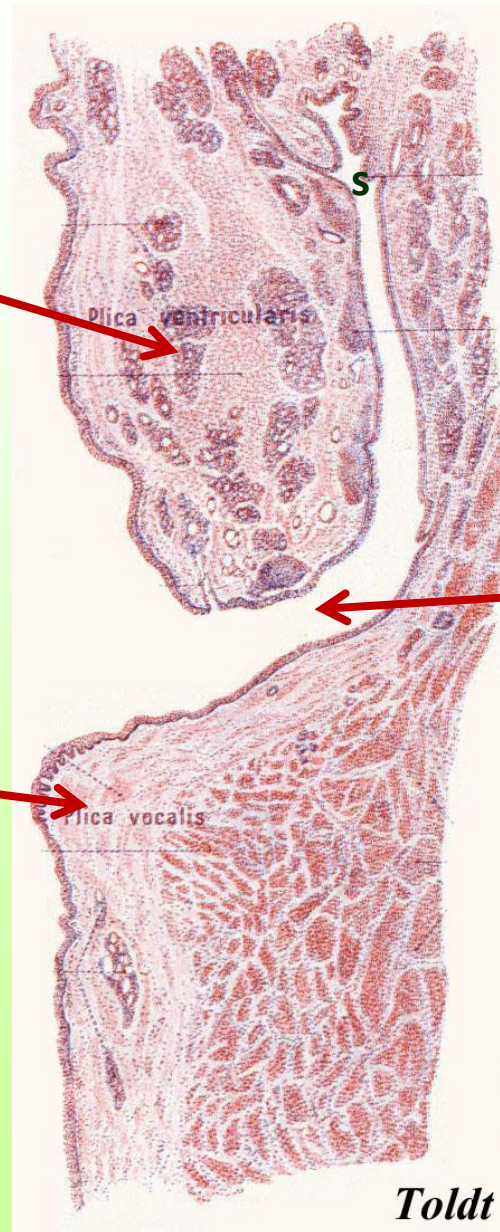
Loose CT (! *Glottis edema* !)
seromucous glands
Quadrangular Membrane

MALT or solitary lymph follicles

VOCAL FOLDS

Stratified non-keratinizing squamous epithelium

Strictly adherent to the vocal folds
No glands
Large muscle content (vocalis m)
Triangular Membrane



Saccule

Extension of the ventricle
(acts as an amplifier in
phonation, vocalization)

VENTRICLE

Toldt